

Hamed Barzamini — ML/AI Engineer

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Summary

ML Engineer building AI-enabled perception systems and evaluation frameworks at scale. Developed black-box evaluation systems achieving 95% F2-score on perception tasks. Experienced in transformers, diffusion models, and LLM-driven semantic specifications. Specialized in dataset quality, concept drift detection, and safety-critical autonomous driving systems.

Experience

Northern Illinois University — Research Assistant / ML Systems Engineer *May 2021 – Present*

- Reduced **AI workflow processing time by 40%** by optimizing pipelines using Intel oneDNN and oneDAL, enabling faster model training and evaluation cycles.
- Identified **80% dataset deficiencies** in pedestrian variations by engineering multi-level semantic web (PyTorch, transformers), leading to **4.9% model accuracy improvement** through targeted data augmentation.
- Improved **model generalizability by 4.9%** (F2-score) using LLM-driven semantic specifications and diffusion-based generative AI (PyTorch, Transformers), reducing model bias by **7.3%** in perception systems.
- Achieved **F2-scores of 95% for pedestrians and 72% for aircraft** by implementing black-box evaluation framework with XAI techniques (Python, PyTorch), enabling systematic benchmarking of perception systems.
- Improved **dataset alignment by 25%** with real-world specifications through systematic dataset evaluation framework using domain-driven embedding techniques, leading to targeted improvements in training data quality.
- Built domain shift detection models using adaptive feature mapping, improving AI robustness against concept drift and changing conditions.

General Motors — Research & Development Intern *Jan 2024 – May 2024*

- Improved **alignment of autonomous driving datasets by 35%** with ODD specifications by developing AI-driven framework using prompt learning and RAG-based knowledge graphs, ensuring higher fidelity in safety-critical scenarios.
- Evaluated dataset completeness for autonomous driving systems, applying framework to assess real-world alignment and improve safety standards in advanced driver assistance systems (ADAS).
- Presented research insights to GM R&D team on dataset validation methodologies, shaping discussions on ADS safety standards.

Additional Research and Development Projects *Nov 2015 – Apr 2020*

- Reduced **software deployment time by 50%** by automating DevOps pipelines with GitLab CI/CD, enabling faster release cycles and improved team productivity.
- Reduced **issue resolution time by 35%** through automated log parsing and web log analysis tools, improving operational efficiency in production systems.
- Reduced **false alerts by 40%** in operational time-series metrics through real-time anomaly detection system, improving system reliability and reducing noise.
- Increased **behavioral prediction accuracy by 30%** using adaptive user profiling framework with machine learning, enhancing personalization and user engagement.
- Boosted **pedestrian detection speed by 35%** through parallelization techniques, enabling real-time processing in safety-critical applications.
- Minimized **software defects by 20%** with rigorous test-driven development (TDD) practices, improving code quality and maintainability.

Projects

B-AIS: Black-box Evaluation Framework — github.com/hamed/bais — demo.bais.com

- Automated black-box evaluation system for visual perception in AI-enabled software using XAI techniques and systematic benchmarking process (Python, PyTorch).
- Achieved **F2-scores of 95% for pedestrians and 72% for aircraft**, identifying weaknesses in model performance through domain semantics integration.
- Published at ASE 2022; framework enables systematic evaluation of safety-critical perception systems.

CADE: Dataset Benchmark for AI Software — github.com/hamed/cade — demo.cade.com

- Systematic dataset evaluation framework using domain-driven embedding techniques (PyTorch, transformers) to assess dataset alignment with real-world specifications.
- Identified **25% deficiencies** in dataset alignment, leading to targeted data augmentation strategies that improved model accuracy by **4.9%**.
- Published at RE 2022; framework is now used for evaluating dataset requirements in ML-based software systems.

V-REX: Requirements Engineering Framework — github.com/hamed/vrex — demo.vrex.com

- LLM-driven requirements engineering framework for AI-enabled perception systems leveraging semantic specifications and diffusion-based generative AI (PyTorch, Transformers).
- Reduced **model bias by 7.3%** and improved dataset completeness, leading to **4.9% accuracy increase** in perception systems.
- In progress (expected 2026); framework transforms vague domain requirements into operational levers for systematic data quality improvement.

ODD Specification Framework for Autonomous Driving — github.com/hamed/odd-framework — demo.odd.com

- AI-driven framework using prompt learning and RAG-based knowledge graphs to specify operational design domains (ODD) in autonomous driving.
- Improved **alignment of autonomous driving datasets by 35%** with ODD specifications, ensuring higher fidelity in safety-critical scenarios.
- Published at RE 2025; framework enhances dataset validation and system robustness for ADAS applications.

Education

Ph.D. in Computer Science, Northern Illinois University 2020 – Present
GPA: 3.83/4.0. Dissertation: *V-REX: Unified Semantics-Driven Requirements Engineering Framework* (In Progress)

Master of Science in Computer Science, Shahid Beheshti University 2012 – 2015
GPA: 3.45/4.0. Thesis: *AMI Theft Detection: A Data-Driven Energy Loss View*

Bachelor of Science in Computer Science, University of Mazandaran 2008 – 2012
GPA: 3.81/4.0, Graduated with highest GPA

Technical Skills

- **Languages:** Python, C++, Java, JavaScript, SQL, C
- **ML/AI:** PyTorch, TensorFlow, Transformers, Diffusion Models, Neural Networks, Deep Learning, NLP, Generative AI, Large Language Models (LLMs)
- **Frameworks & Libraries:** SciPy, OpenCV, Observable, D3.js
- **Data Analytics:** Apache, MySQL, Pandas
- **Cloud/DevOps:** Docker, Git, Intel AI Analytics Toolkit, Intel oneAPI
- **Software Engineering:** AI-enabled Systems Design, Requirements Engineering, Software Development
- **Algorithms:** Genetic Algorithms, Monte-Carlo Tree Search, Dynamic Programming, Graph Algorithms, Greedy Algorithms